Ömer Aydın

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About Me

Product-minded Senior Software Engineer focused on backend systems and agentic AI. I build distributed backend services, APIs, and data pipelines, and develop agentic workflows with an emphasis on evaluation and observability. Previously a founding engineer on nmaistro.ai, where I built the MCP integration layer from scratch. I enjoy solving complex system challenges, building high-performance software, and taking existing systems apart to understand how they work.

Experience

Senior Software Engineer Feb. 2026 – Present
Tangany GmbH *Munich, Bavaria, DE*

- Proposed and built an agentic AI system to automate AML source-of-funds and source-of-wealth checks after Tangany onboarded Nexo, a large digital asset platform. Designed it to handle the extra compliance work from rapid customer growth without hiring dozens of additional AML analysts.
- Implemented its multi-agent pipeline with **Vercel AI SDK** and Azure-hosted models in **Microsoft Foundry**. The agents use Mistral and Azure Document Intelligence for **OCR** and structured extraction, then call tools to assess customer evidence using blockchain transactions, historical prices, and AML rules.
- Created the system's LLM evaluation workflow in **Langfuse**, using benchmark datasets and experiments to test extraction and assessment, catch regressions, and compare prompt versions. Used experiment results to improve prompts and traces to debug individual agent runs.
- Defined the analyst workflow and built a **React/TypeScript** dashboard backed by PostgreSQL for reviewing automated assessments and investigating cases through AI chat.
- Contributed to custody backend services and REST APIs with **Node.js, TypeScript, and Nest.js**, including automated testing and Azure CI/CD.

Senior Software Engineer Jan. 2025 – Feb. 2026
NewMind AI *İstanbul, TR*

- Founding engineer on nmaistro.ai, an AI agent orchestration platform that turns natural-language requests into workflows across connected tools, producing documents, dashboards, and other dynamic artifacts. Architected the backend microservices, contributed to the AI agent runtime, and designed and built the entire MCP integration layer from scratch, supporting 200+ integrations. Partnered on system design and tech-stack decisions.
- Led R&D and backend architecture across AI-driven legal tech products, guiding multiple teams through complex system design, optimization, and implementation.
- Architected distributed microservices and data pipelines that significantly improved system scalability, resilience, and developer productivity across the platform.
- Designed and built a custom Elasticsearch access-control plugin in Java & Go, scaling to millions of permission checks across millions of legal documents, leveraging RoaringBitmaps and low-level engine integrations for sub-millisecond access validation.
- Developed a modular ETL and knowledge graph builder in Node.js/TypeScript, transforming diverse legal datasets into large-scale Neo4j graph structures with automated schema generation and CLI-driven workflows.
- Mentored junior developers, established engineering standards, and drove internal tooling and automation that accelerates delivery across R&D teams.

Software Engineer Feb. 2022 – Dec. 2024
NewMind AI *İstanbul, TR*

- Contributed to the core development of AI-powered legal tech platforms, building scalable backend systems and integrating complex data pipelines for global law firms and enterprises.
- Designed and implemented microservices in Node.js/Nest.js and Go, using Kafka, Elasticsearch, and GraphQL to enable efficient, event-driven communication and data retrieval at scale.
- Led major refactors and modernizations, including the migration from legacy PHP monoliths to microservice architectures, cutting load times and improving maintainability across multiple products.

- Enhanced system performance through advanced database optimizations in PostgreSQL, MongoDB, and Redis, reducing query times and improving throughput.
- Collaborated closely with frontend and mobile teams to deliver end-to-end product features, contributing to improved UX and faster delivery cycles.
- Built and deployed a cross-platform Flutter time-tracking app for lawyers and staff with push notifications, working closely with backend teams to refine APIs and integrate backend services.

Software Developer Sep. 2017 – Feb. 2022
Freelance *Worldwide*

- Delivered custom Android applications for international clients using Java and Kotlin, focusing on performance, clean architecture, and long-term maintainability.
- Built and deployed backend services with Node.js and PHP, managing server operations, databases, and scaling for production environments.
- Developed automation scripts and data tools in Python, including web scrapers and API integrations to streamline client workflows.

Volunteering

Backend & Python Teacher Mar. 2026 – Present
ReDI School of Digital Integration *Munich, Bavaria, DE*

- Teach backend development and Python through practical examples drawn from industry experience in software engineering, automation, and API integrations. Support learners in developing problem-solving skills and the confidence to pursue careers in technology.

Skills

Languages & Frameworks: Node.js, Nest.js, JavaScript, TypeScript, Java, Go, Python, Kotlin, Android, Flutter, React Native, Bun, Hono, Fastify, tRPC, React, Tailwind
Databases & Queues: MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, Couchbase, Kafka, ClickHouse, CockroachDB, RabbitMQ, Neo4j, Memgraph
APIs & Protocols: REST, GraphQL, gRPC
Agentic AI: Multi-Agent Orchestration, Tool Calling, Prompt Engineering, LLM Evaluation & Observability, RAG, Knowledge Graphs, OCR & Structured Extraction
AI Tools & Protocols: Vercel AI SDK, Langfuse, Agno, Mastra, LangChain, MCP, A2A, AG-UI
Developer Tools: Docker, Kubernetes, Git, Linux, GitLab CI/CD, GitHub Actions, Vite
Cloud & Services: Azure, GCP, Cloudflare, Firebase
Spoken Languages: English (Fluent), Turkish (Native)

Education

İstanbul University - Cerrahpaşa Graduate
Bachelor's Degree in Computer Engineering

Projects

Dexter | *Java, Kotlin, Android* Sep. 2021 – Present

- Developed an open-source DEX/APK editor and reverse-engineering platform for Android, earning 150+ GitHub stars from the security research community.
- Used Smali and Dexlib2 for low-level Android bytecode manipulation and analysis, supporting security research and vulnerability assessment.
- Integrated five industry-standard decompilers (JADX, Fernflower, CFR, JD-Core, Procyon) with custom optimizations for on-device Java decompilation and real-time code analysis.
- Implemented syntax highlighting, Smali navigation, and real-time error detection, with an extensible architecture for future APK reconstruction and Java bytecode assembly.

Personal Website | *TypeScript, Bun, Hono, Vite, Cloudflare* Mar. 2024 – Present

- Developed a high-performance static site generator in TypeScript using Bun and Hono, with Vite for build optimization.
- Engineered a type-safe content management system with structured TypeScript models, compiling Markdown content into static HTML.
- Built a GitHub Actions CI/CD pipeline for automated testing, builds, and deployment to Cloudflare Pages.